Write a program to implement framing: i. Fixed framing, ii. Variable framings

Aim : To write a C program to implement **Fixed Framing**, where the input data is divided into frames of a fixed size.

Description

Frames can be of **fixed or variable size**. In **fixed-size framing**, there is **no need for defining the boundaries of the frames**; the **frame size itself acts as the delimiter**. An example of this type of framing is the **ATM (Asynchronous Transfer Mode) wide-area network**, which uses fixed-size frames called **cells**.

Algorithm

1. Read the input data from the user.
2. Read the frame size.
3. Calculate the length of the data.
4. Divide the data into frames of the specified size.
5. Display each frame with its frame number.
6. Stop when all data has been divided into frames.

Program Fixed framing:

```
#include <stdio.h>
#include <string.h>

int main()
{
    char data[100];
    int frameSize, len, i, frameNo = 1;

    printf("Enter the data: ");
    scanf("%s", data);

    printf("Enter frame size: ");
    scanf("%d", &frameSize);

    len = strlen(data);

    printf("\nFixed Frames:\n");

    for(i = 0; i < len; i += frameSize)
    {
        printf("Frame %d : ", frameNo++);

        for(int j = 0; j < frameSize && (i + j) < len; j++)
        {
            printf("%c", data[i + j]);
        }

        printf("\n");
    }

    return 0;
}
```

Sample Output

```
Enter the data: COMPUTERNETWORK
Enter frame size: 4

Fixed Frames:

Frame 1 : COMP
Frame 2 : UTER
Frame 3 : NETW
Frame 4 : ORK
```

Aim : To write a C program to implement **Variable Framing**, where the input data is divided into frames of variable sizes specified by the user.

Description

Variable framing is a framing technique in which **frames can have different lengths** depending on the amount of data or user-defined frame size. Unlike fixed framing, each frame does not have to be of equal size. This method provides flexibility and is useful in protocols where data packets vary in length. The program repeatedly accepts a frame size from the user and divides the input data accordingly until all characters are transmitted.

In **variable-size framing**, we need a way to **define the end of one frame and the beginning of the next frame**.Historically, two approaches were used for this purpose:

- **Character-oriented approach**
- **Bit-oriented approach**

Algorithm

1. Read the input data from the user.
2. Calculate the length of the data.
3. Initialize the starting position of the data.
4. Read the frame size from the user.
5. Extract and display the specified number of characters as one frame.
6. Repeat Steps 4 and 5 until the entire data is divided into frames.
7. Stop when all characters have been transmitted

Program variable length frame:

```
#include <stdio.h>

#include <string.h>

int main()

{

   char data[100];

   int len, i = 0, frameSize;

   printf("Enter the data: ");

   scanf("%s", data);
```

```
    len = strlen(data);
    printf("\nVariable Frames:\n");
    while(i < len)
    {
        printf("Enter frame size: ");
        scanf("%d", &frameSize);
        printf("Frame: ");
        for(int j = 0; j < frameSize && i < len; j++, i++)
        {
            printf("%c", data[i]);
        }
        printf("\n");
    }
    return 0;
}
```

Output

```
Enter the data: COMPUTERNETWORK
Variable Frames:
Enter frame size: 3
Frame: COM
Enter frame size: 5
Frame: PUTER
Enter frame size: 4
Frame: NETW
Enter frame size: 5
Frame: ORK
```